the same thing as a nondegenerate invariant sesquilinear form. So one can use a unitary structure on V to define an isomorphism $\overline{V} \to V^*$.

Theorem 4.6.3. *A finite dimensional unitary representation V of any group G is completely reducible.*

Proof. Let W be a subrepresentation of V. Let $W^\perp$ be the orthogonal complement of W in V under the Hermitian inner product. Then $W^\perp$ is a subrepresentation of V, and $V = W \oplus W^\perp$. This implies that V is completely reducible. $\square$

Theorems 4.6.2 and 4.6.3 imply Maschke's theorem for complex representations (Theorem 4.1.1). Thus, we have obtained a new proof of this theorem over the field of complex numbers.

Remark 4.6.4. Theorem 4.6.3 shows that for infinite groups G, a finite dimensional representation may fail to admit a unitary structure (as there exist finite dimensional representations, e.g., for $G = \mathbb{Z}$, which are indecomposable but not irreducible).

4.7. Orthogonality of matrix elements

Let V be an irreducible representation of a finite group G, and let $v_1, v_2, \ldots, v_n$ be an orthonormal basis of V under the invariant Hermitian form. The matrix elements of V are $t_{ij}^V(x) = (\rho_V(x)v_i, v_j)$.

Proposition 4.7.1. *(i) Matrix elements of nonisomorphic irreducible representations are orthogonal in $F(G, \mathbb{C})$ under the form $(f, g) = \frac{1}{|G|}\sum_{x\in G} f(x)\overline{g(x)}$.*

(ii) One has $(t_{ij}^V, t_{i'j'}^V) = \delta_{ii'}\delta_{jj'} \cdot \frac{1}{\dim V}$.

Thus, matrix elements of irreducible representations of G form an orthogonal basis of $F(G, \mathbb{C})$.

Proof. Let V and W be two irreducible representations of G. Take $\{v_i\}$ to be an orthonormal basis of V and $\{w_i\}$ to be an orthonormal basis of W under their positive definite invariant Hermitian forms. Let $w_i^* \in W^*$ be the linear function on W defined by taking the inner product with w_i: $w_i^*(u) = (u, w_i)$. Then for $x \in G$ we have